

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD materials chemist with 6+ years in analytical characterization, stability/degradation analysis, thermal/spectroscopic methods, and DOE-guided process-property studies. Brings a materials-science lens to formulation, particle/biophysical characterization, technical reports, and cross-functional CMC problem solving while transparently bridging into biologics-specific methods.

CORE COMPETENCIES

Materials & Biophysical Characterization

Formulation Stability

IR / DSC / Thermal Analysis

Particle/Method Development Mindset

DOE/SPC

Technical Reports

Cross-Functional CMC

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided polymer process-development studies, defining analytical readouts, process windows, reproducibility criteria, and SOP-ready specifications for robust material quality.
- Connect formulation/process variables with FT-IR, thermal-analysis, and spectroscopy data to support stability interpretation, troubleshooting, and technical recommendations.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Characterized 15+ functional material variants using UV-Vis-NIR, FT-IR, cyclic voltammetry, spectroelectrochemistry, NMR, GPC, TGA, DSC, and MS-informed analysis.
- Mapped degradation and stability failure modes through hypothesis-driven experiments, orthogonal characterization, and rigorous technical documentation.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Developed molecular thin-film coating processes and linked CVD/surface-preparation variables with morphology, interface quality, and material performance.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Optimized CVD polymer thin-film deposition with DOE strategies and in-situ QCM-D monitoring to quantify growth kinetics and improve reproducibility.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30% and strengthened data integrity.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Stability and Degradation Characterization Chemistry of Materials 2025

Mapped material stability and degradation mechanisms using orthogonal spectroscopy, electrochemistry, and thermal analysis.

FT-IR, UV-Vis-NIR, CV, DSC/TGA, NMR, GPC, MS

Process-Property Method Development Current

Built analytical readouts and process windows that connect material attributes to reproducible specifications.

DOE, SPC, FT-IR, thermal analysis, SOPs

Instrumentation and Lab Operations Leadership

Managed calibration, SOP, safety, and equipment systems supporting reliable multi-user characterization workflows.

Instrument uptime, documentation, training, safety

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Biophysical/Materials Characterization: FT-IR/IR, UV-Vis-NIR, spectroelectrochemistry, TGA, DSC, NMR, GPC/SEC, QCM-D, GC-MS/LC-MS exposure; DLS/CD/AUC familiarity to be developed

Stability & Formulation Mindset: Material degradation, aggregation/stability concepts, process-composition-property relationships, particle-method development mindset, troubleshooting

Process & Data Methods: DOE, SPC, JMP/basic, Excel, Python/basic, reproducibility studies, method comparison, technical reports

Execution: SOPs, lab notebooks, instrument troubleshooting, calibration tracking, cross-functional communication, publication-quality writing

JOURNAL PUBLICATIONS

- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.
- Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.
- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)